



Virtual SOC Capstone

Charter

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1. Project Vision

Create a virtual Security Operations Center (SOC) environment on Miami University's OpenStack private cloud to support teaching and demonstration of cybersecurity concepts, including attack execution, detection, and incident analysis.

2. Project Mission

Enable students and instructors to demonstrate cybersecurity activities in a fully isolated, controlled environment by:

- Simulating attacks using tools such as Metasploit and Kali Linux
- Analyzing system and network evidence generated by attacks
- Investigating simulated security incidents using logs from integrated Endpoint Detection and Response, Firewall, Network Intrusion Detection Systems, and Security Information and Event Manager tools
- Visualizing security activity through the SIEM to provide meaningful dashboards for teaching and demos

3. Success Criteria

- **Attack Simulation:** We should be able to successfully demonstrate attacks on all segments of our network and view the evidence of said attacks

- **Network Isolation:** All systems should be on our allocated Openstack domain and should have no connection to Miami University's campus network during attack simulations
- **Detection of Attacks:** All intrusion detection systems and sensors should properly capture logs and generate alerts so they can be analyzed for teaching purposes
- **Automation of Deployment:** The entire environment should be fully deployable via script
- **Documentation and Walkthroughs:** Step-by-step instructions are provided for setup, usage, and teardown of the SOC
- **Visualization and Teaching Support:** Dashboards provide meaningful, real-time visualizations of network activity. Instructor station (DSB 365) supports multiple monitors/projector views for live teaching demonstrations.

4. Constraints, Risks, and Issues

- **Schedule Slippage:** There is a chance that the team gets off schedule due to reasons beyond their control. Such as but not limited to, technical constraints, class schedules not allowing for all members to meet, or client schedules not allowing to meet and get next requirements.
- **Misconfiguration of Network:** If the network is misconfigured in set up, then there is the possibility that the internal network is exposed to external attacks that are unwanted in the simulation

- **Technical Inexperience:** Since most of the technology is new to team members there is the possibility that inexperience will lead to constraints and difficulties that can not be predicted
- **Truck Factor:** If one of the team members is unable to contribute to the project for an indefinite period, will the project still be able to move forward.